AJKD Atlas of Renal Pathology: Anti–Tubular Basement Membrane Antibody Disease

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Clinical and Pathologic Features

Anti–tubular basement membrane (anti-TBM) antibody disease is rare and presents with acute or chronic kidney injury. Patients may be of any age and usually have polyuria and polydipsia. Microhematuria and proteinuria (sometimes nephrotic-range) are also seen.

Light microscopy: There is an interstitial mononuclear infiltrate with lymphocytes, plasma cells, and macrophages, with occasional neutrophils. Variable interstitial eosinophils can also be present with tubulitis with associated acute tubular injury and interstitial edema, rarely with interstitial giant cells. Glomeruli and arteries show nonspecific changes. The extent of interstitial fibrosis and tubular atrophy varies depending on the chronicity of anti-TBM disease at the time of biopsy.

Immunofluorescence microscopy: There is strong diffuse linear staining of TBMs for IgG, with variable C3 staining.

Electron microscopy: No deposits are seen.

Etiology/Pathogenesis

Primary anti-TBM disease is rare and is related to an autoantibody to a protein called tubulointerstitial nephritis antigen, which is only expressed in the kidney. This antigen is a 58-kDa noncollagenous protein found in the proximal TBM. It is a regulatory protein in tubulogenesis that interacts with type IV collagen, laminin, and integrins. In some patients, drug exposure may have triggered autoantibody formation. Serum anti-TBM antibodies are detected.

Differential Diagnosis

Anti-TBM disease can be associated with membranous nephropathy in children. Anti-TBM disease can occur in the kidney allograft in recipients who lacked the antigen in their native kidneys. The autoantibody in anti–glomerular basement membrane (GBM) disease can crossreact with basement membranes of distal tubules with resulting focal linear TBM staining, in addition to the characteristic strong linear GBM staining with accompanying glomerular crescentic injury seen in anti–GBM antibody–mediated disease.

Key Diagnostic Features

- Interstitial mononuclear infiltrate with tubulitis and tubular injury
- Strong linear IgG staining along TBMs by immunofluorescence
- Anti-TBM antibodies detected in serum

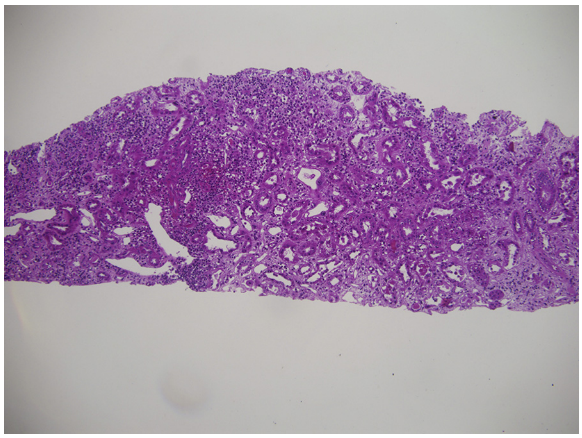

Figure 1. Anti–tubular basement membrane disease with diffuse interstitial lymphocytic infiltrate, with occasional plasma cells and eosinophils associated with tubulitis and mild interstitial edema (hematoxylin and eosin stain).

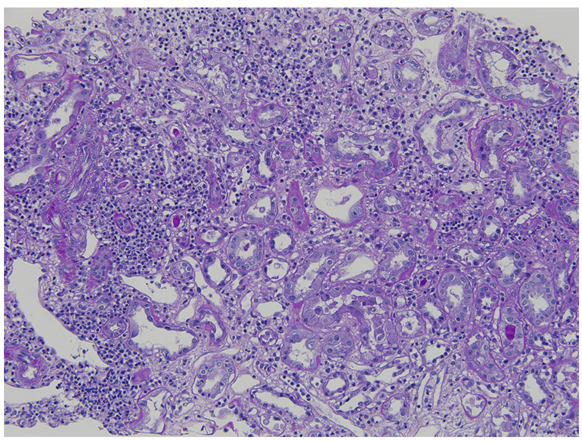

Figure 2. Anti–tubular basement membrane disease with interstitial lymphocytic infiltrate, with rare eosinophils associated with tubulitis and mild interstitial edema (periodic acid–Schiff stain).

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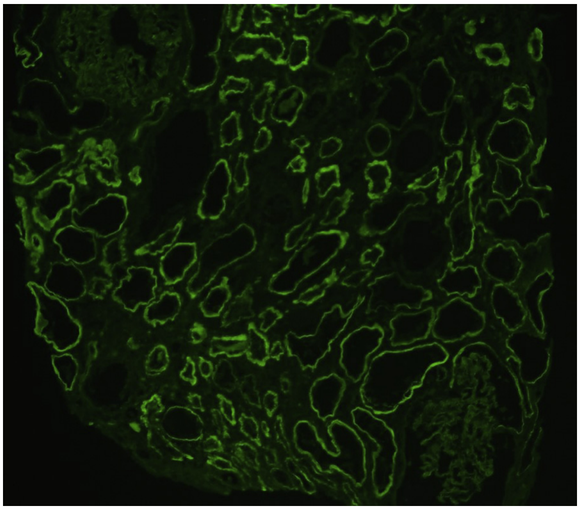

Figure 3. Anti–tubular basement membrane disease with strong smooth linear staining of tubular basement membranes without glomerular basement staining with antibody to IgG (immunofluorescence microscopy).